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Answer: D

Gain in GPE = Loss in KE

$$mgr(1 + \cos \theta) = \frac{1}{2}m(v_i^2 - v_f^2)$$

$$gr(1 + \cos \theta) = \frac{1}{2}(v_i^2 - v_f^2)$$

$$(9.81)(7.7)(1 + \cos 30^\circ) = \frac{1}{2}(25^2 - v_f^2)$$

$$V_f = 18.5 \text{ m s}^{-1}$$

Assume rod in tension

$$mg + F = \frac{mv^2}{r}$$

$$(3.5)(9.81) + F = \frac{(3.5)(18.5)^2}{7.7}$$

$$F = 121 \text{ N} \quad (\text{tension})$$

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Answer: B